

Lorenzo Stacchio, PhD

Extended Reality, Generative Artificial Intelligence, Human-Centered Interactive Systems

Department of Political Science, Communication and International Relations, University of Macerata |
lorenzo.stacchio@unimc.it | [ORCID: 0000-0002-9341-7651](https://orcid.org/0000-0002-9341-7651) | [Scopus](#) | [Web of Science](#)

Academic Profile

Lorenzo is a Computer scientist and researcher working at the intersection of Extended Reality (XR), Artificial Intelligence (AI), Computer Vision, Human-Computer Interaction, Generative AI, Digital Twins and Human-Centered Evaluation. The research agenda is centered on *Extended Artificial Intelligence* (XRAI): AI systems embedded in immersive environments to support human activity, creativity, training, cultural heritage, education, industrial workflows and interactive decision-making.

Lorenzo is a Post-Doc researcher in Computer Science and Engineering at the University of Macerata, Department of Political Science, Communication and International Relations (SPOCRI), and is a member of the Vision Robotics and Artificial Intelligence (VRAI) Lab. His profile integrates research, teaching, project design, supervision, dissemination, scientific service and technology transfer.

Keywords.

Extended Reality

Generative AI

Multimodal LLMs

Human-in-the-Loop

Neural Rendering

Human Computer Interaction

Creative Industries

Cultural Heritage

Distinctive Contribution

- **Scientific integration:** design of AI/XR pipelines where perception, generation, spatial interaction and user evaluation are treated as a unified system rather than as isolated components.
- **Applied interdisciplinarity:** deployment of computer vision, generative AI and immersive interaction in fashion, cultural heritage, education, sport, retail, manufacturing and communication contexts.
- **Teaching transfer:** translation of research prototypes into modular learning experiences for doctoral, master's, bachelor's and professional-development audiences.
- **Evaluation culture:** emphasis on measurable learning outcomes, usability, cognitive load, social presence, acceptance, ethics and responsible AI/XR design.

Positions and Education

Academic and Professional Positions

Feb. 2026 – present

Research Fellow

University of Macerata, SPOCRI Department

Research, teaching, organization and technology-transfer activities in Computer Science and Information Engineering (IINF-05/A). Research area: consumer behavior analysis in physical retail through top-view RGB-D cameras. Supervisor: Prof. Emanuele Frontoni.

Feb. 2025 – Jan. 2026

Research Fellow

University of Macerata, SPOCRI Department

Research, teaching, organization and technology-transfer activities in Computer Science and Information Engineering (IINF-05/A). Research area: consumer behavior analysis in physical retail through top-view RGB-D cameras. Supervisor: Prof. Marina Paolanti.

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Research, teaching, organization and technology-transfer activities in Computer Science and Information Engineering (IINF-05/A). Research area: consumer behavior analysis in physical retail through top-view RGB-D cameras. Supervisor: Prof. Marina Paolanti.

A.Y. 2025/26	Adjunct Professor Università Politecnica delle Marche <i>Course holder for Advanced Videogame and Virtual Reality Systems Programming, 9 CFU, 72 hours, B.Sc. in Information Engineering for Videogames and Virtual Reality. Topics: advanced programming, game engines, VR systems, and applied interactive projects.</i>
Oct. 2024; Oct. 2025	Doctoral Module Lecturer University of Milano-Bicocca <i>Lecturer for Deep Learning and Computer Vision for Business, 2 CFU, PhD program in Economics, Statistics and Data Science.</i>
Feb. 2023 – May 2023	Visiting Researcher TU Graz University <i>Research period on diffusion models for temporally coherent rendering generation. Supervisor: Prof. Denis Kalkofen.</i>

Education

Nov. 2020 – Jan. 2024	PhD in Computer Science University of Bologna <i>Doctoral research in Computer Science (INF/01) within the PhD program in Quality of Life Sciences, XXXVI cycle. Dissertation: Entangling artificial intelligence with extended reality paradigms for human activity support. Advisor: Prof. Gustavo Marfia; Co-advisor: Prof. Giuseppe Lisanti.</i>
Sept. 2018 – Oct. 2020	M.Sc. in Computer Science, 110/110 cum laude University of Bologna <i>Software Techniques specialization.</i>
Sept. 2015 – July 2018	B.Sc. in Computer Science and Technology, 110/110 cum laude University of Camerino
Sept. 2010 – July 2015	Technical High School Diploma 100/100 Istituto Tecnico Commerciale Alberico Gentili, Macerata <i>Specialization in information systems for business.</i>

Teaching Portfolio

Official Courses

A.Y. 2025/26	Advanced Videogame and Virtual Reality Systems Programming Università Politecnica delle Marche <i>9 CFU (72 hours) course in the B.Sc. program in Information Engineering for Videogames and Virtual Reality. Topics: advanced programming, game engines, VR systems, and applied interactive projects.</i>
A.Y. 2023/24–2024/25–2025/26	Deep Learning and Computer Vision for Business II University of Milano-Bicocca <i>2 CFU (10 hours) doctoral module in the PhD program in Economics, Statistics and Data Science. Topics: computer vision pipelines, deep learning models, and business-oriented applications.</i>

Teaching Support, Tutoring and Student Supervision

- Co-supervisor of 9 bachelor/master theses at the University of Bologna, University of Macerata and E-Campus University, covering computer science, fashion studies, communication and engineering topics.
- Teaching-support and examination-board activities at the University of Macerata for courses in multimedia informatics, AI, social media, XR/metaverse design, web design, AI for communication and GeoAI.
- Teaching-support activities at the University of Bologna for *Laboratory of Virtual and Augmented Reality*, *Information Technologies for Fashion*, *E-Commerce for Fashion*, and *Data Science and Immersive Technologies for Fashion E-Commerce*.
- Teaching-support activities at Università Politecnica delle Marche for *Computer Graphics and Multimedia*.
- Tutor for *Algorithms and Data Structures* at the University of Bologna and laboratory tutor for VARLAB activities.

Seminars, Lifelong Learning and Third Mission

- *Artificial Intelligence for Fashion*, 30-hour seminar cycle, University of Bologna.
- *An Introduction to Computer Science for Fashion*, 10-hour crash course, University of Bologna.
- STEM4Humanities Erasmus+ teaching activities on STEM/STEAM skills for humanities professors.
- PNRR training activities on AI in the classroom, metaverse and inclusive technologies.
- Teacher-training activity *AI and XR for innovation in practical education and training*.
- Big Data laboratory for secondary schools, co-funded by Regione Emilia-Romagna within the open laboratories program.

Research Profile

The research activity focuses on the convergence of Extended Reality and Artificial Intelligence to develop systems that support human activity in interdisciplinary contexts. The work covers immersive experiences in VR, AR and MR, machine-learning and deep-learning architectures, multimodal perception, generative AI, digital twins and human-centered evaluation. This agenda is framed through *Extended Artificial Intelligence* (XRAI): an integrated approach in which AI becomes spatial, embodied, context-aware and interactive.

Theoretical Axes

- **Computer vision and multimodal recognition:** supervised and unsupervised models for visual classification, cultural-relationship extraction, object recognition and situated visualization.
- **Generative AI and XR content creation:** diffusion models, multimodal LLMs, large reconstruction models and neural rendering to support coherent immersive content generation.
- **Digital twins and collaborative XR:** interactive digital twins for instruction learning, annotation, simulation, manufacturing, urban planning and procedural support.
- **Human-centered interaction:** evaluation of presence, engagement, cognitive load, emotional response, usability and trust in immersive environments.
- **Responsible XRAI:** ethical frameworks addressing bias, privacy, transparency, explainability and safe deployment in AI/XR systems.

Application Domains

Domain	Research contribution	Representative outcomes
Creative industries and fashion	Immersive x-commerce, generative content design, neural rendering and cross-reality collaboration.	Voice-augmented VR shopping; fashion neural rendering; CRoCs-XR; CreAIXR.
Cultural heritage and digital humanities	AI-based extraction, annotation and immersive exploration of historical, visual and performative heritage.	IMAGO-based socio-historical photo analysis; DanXe for dance heritage; AR family photo albums.

Education and training	XR and AI systems for procedural learning, embodied tutoring, practical simulation and assessment.	Rubik's Cube digital-twin instruction; virtual humans for educational scenarios; XR-based workout guidance.
Industry and manufacturing	Mixed-reality instruction, annotation, Lean Manufacturing and simulation transfer.	Annholotator; Unity-VRLines; Flying in XR middleware.
Retail and behavior understanding	Human behavior analysis through RGB-D cameras, object-level tracking and foundation-model-oriented retail analytics.	Top-view retail behavior analysis; ego/exo view tracking; physical retail AI workshop contributions.

Current Research Direction

A natural evolution of this profile is the design of agentic and multimodal XR systems: AI agents able to perceive spatial context, reason over user goals, generate or retrieve content, select tools, and act through immersive interfaces. This direction connects the candidate's prior work on XR, generative AI, digital twins, multimodal LLMs and human-in-the-loop evaluation into a unified research line for future AI-enhanced spatial computing.

Projects, Scientific Service and Awards

Research Projects and Responsibilities

2024–2025	STEM Skills for Humanities Erasmus+ KA220 – Cooperation Partnerships in Higher Education <i>Responsible for the design and implementation of innovative training pathways in Work Package 2, supporting STEM/STEAM upskilling for humanities professors.</i>
2025–present	AGRITECH.EU – Digital Agriculture for Sustainable Development Digital Europe Programme <i>Responsible for digital content and training-program development in Work Packages 2 and 3, including AI/XR-oriented educational resources for digital agriculture.</i>
2024–present	Regional and industrial innovation projects Marche Region and industrial partners <i>Participation in CAPPELLAI, PASTA, MuseAI, IDEA and G-NEXT, focusing on AI for creative industries, production, museum interaction, defect detection and sustainable product/service innovation.</i>

Scientific Groups and Collaborations

- Member of VRAI - Vision, Robotics and Artificial Intelligence Lab, connected with the University of Macerata and Università Politecnica delle Marche (since 2024).
- Member and collaborator of VARLAB - Virtual and Augmented Reality Laboratory, University of Bologna.
- Collaborator of CFC - International Research Center Culture Fashion Communication, University of Bologna.
- Member of XRItalia, the Italian association for research in Extended Reality from 2026.
- Member of CVPL, the Italian association for research in Computer Vision, Pattern Recognition, and Machine Learning from 2024.

Scientific Organization and Invited Talks

- Principal Organizer of the *1st International Workshop on Agentic AI for Extended Reality (XRAG)*, ISMAR 2026 (A* on Core).
- Principal Organizer of the *1st International Workshop on Humans in the eXtended Artificial Intelligence Loop (HUARL)*, ICIAP 2025.
- Session Chair of the International Workshop on Artificial Intelligence and Creativity (CREAI), ECAI 2024.
- Co-chair of the special session *FAME: Fashion for Metaverse*, AIVR 2024.
- Organizer of the conference tutorial *Fashion in the Metaverse: Technologies, Applications, and Opportunities*, IFIP-ICEC 2023.
- Invited international speaker at the 2nd Physical Retail AI Workshop, WACV 2025, with a talk on Extended Artificial Intelligence in physical retail.

Editorial and Reviewing Activities

- Associate Editor of *The Visual Computer* from February 2025.
- Reviewer for international journals published by IEEE, ACM, Springer, Elsevier, Taylor & Francis, MDPI, and Frontiers.
- Program Committee member and reviewer for conferences and workshops, including ISMAR, IEEE VR, AAAI, ACM MM, ICC, ECCV, IROS, ISRR and ICIAP.

Awards and Recognition

- Best Poster at VisMac2020 Doctoral School for the paper *IMAGO: a family photo album dataset for a socio-historical analysis of the twentieth century*.
- Winner of the best project competition at the international Summer School *XR and AI for enhancing cultural and territorial heritage*, Matera, 2022.
- Best Paper at the 32nd IEEE Conference on Virtual Reality and 3D User Interfaces for *MineVRA: Exploring the Role of Generative AI-Driven Content Development in XR Environments through a Context-Aware Approach*.

Selected Publications

[†] Co-first authors or same contributions.

Journal Papers

- [J17] Balloni, E., **Stacchio, L.**, Paolanti, M., Zingaretti, P., & Pierdicca, R. (in press). Orchestrating Generative AI Paradigms with Human-In-The-Loop for 3D Generation. *IEEE Transactions on Visualization and Computer Graphics*.
- [J16] Sirocchi, C.[†], **Stacchio, L.**[†], Migliorelli, L., Frontoni, E., & Mancini, A. (in press). XaRNold: Assessing the role of virtual humans in immersive fitness environments with aligned flat, multi-view, and 3D parametric visual instruction guidance. *Virtual Reality*.
- [J15] Saraceni, E., Migliorelli, L., Frontoni, E., & **Stacchio, L.** (2026). Beyond screen time: Analyzing the discrepancy between objective and perceived smartphone usage in adolescents. *Frontiers in Computer Science*, 8.
- [J14] Hajahmadi, S., Cascarano, P., **Stacchio, L.**, Giacchè, A., & Marfia, G. (in press). Voice-augmented extended reality digital twin instruction: The case of the Rubik's Cube. *Virtual Reality*.
- [J13] **Stacchio, L.**, Nepi, L., Paolanti, M., & Pierdicca, R. (2025). RSplitzero: Generalized zero-shot learning in remote sensing across attribute splits with single- and multi-modal representations. *International Journal of Digital Earth*, 18(2), 2551869.
- [J12] Armandi, V., **Stacchio, L.**, Cascarano, P., Hajahmadi, S., Donatiello, L., & Marfia, G. (2025). An augmented outdoor workout system for jogging and calisthenics support. *Frontiers in Virtual Reality*, 6, 1613717.
- [J11] Balloni, E., **Stacchio, L.**, Mancini, A., Frontoni, E., Zingaretti, P., & Paolanti, M. (2025). A neural rendering system for fashion design process. *Engineering Applications of Artificial Intelligence*, 153, 110773.
- [J10] Santarnecchi, E., Balloni, E., Paolanti, M., Frontoni, E., **Stacchio, L.**, Zingaretti, P., & Pierdicca, R. (2025). MineVRA: Exploring the role of generative AI-driven content development in XR environments through a context-aware approach. *IEEE Transactions on Visualization and Computer Graphics*.
- [J9] **Stacchio, L.**, Balloni, E., Gorgoglione, L., Mancini, A., Giovanola, B., Tiribelli, S., & Zingaretti, P. (2024). An ethical framework for trustworthy neural rendering applied in cultural heritage and creative industries. *Frontiers in Computer Science*, 6, 1459807.
- [J8] **Stacchio, L.**, Angeli, A., Lisanti, G., & Marfia, G. (2024). Analyzing cultural relationships visual cues through deep learning models in a cross-dataset setting. *Neural Computing and Applications*, 36(20), 11727–11742.
- [J7] **Stacchio, L.**, Garzarella, S., Cascarano, P., De Filippo, A., Cervellati, E., & Marfia, G. (2024). DanXe: An extended artificial intelligence framework to analyze and promote dance heritage. *Digital Applications in Archaeology and Cultural Heritage*, 33, e00343.
- [J6] Scorolli, C., Grasso, E. N., **Stacchio, L.**, Armandi, V., Matteucci, G., & Marfia, G. (2023). Would you rather come to a tango concert in theater or in VR? Aesthetic emotions and social presence in musical experiences, either live, 2D or 3D. *Computers in Human Behavior*, 149, 107910.
- [J5] Angeli, A., **Stacchio, L.**, Donatiello, L., Giacchè, A., & Marfia, G. (2024). Making paper labels smart for augmented wine recognition. *The Visual Computer*, 40(8), 5519–5531.

- [J4] Morotti, E., **Stacchio, L.**, Donatiello, L., Rocchetti, M., Tarabelli, J., & Marfia, G. (2022). Exploiting fashion x-commerce through the empowerment of voice in the fashion virtual reality arena: Integrating voice assistant and virtual reality technologies for fashion communication. *Virtual Reality*, 26(3), 871–884.
- [J3] **Stacchio, L.**, Angeli, A., Lisanti, G., Calanca, D., & Marfia, G. (2022). Toward a holistic approach to the socio-historical analysis of vernacular photos. *ACM Transactions on Multimedia Computing, Communications and Applications*, 18(3s), 1–23.
- [J2] Bononi, L., Donatiello, L., Longo, D., Massari, M., Montori, F., **Stacchio, L.**, & Marfia, G. (2022). Digital twin collaborative platforms: Applications to humans-in-the-loop crafting of urban areas. *IEEE Consumer Electronics Magazine*, 12(6), 38–46.
- [J1] **Stacchio, L.**, Angeli, A., & Marfia, G. (2022). Empowering digital twins with extended reality collaborations. *Virtual Reality & Intelligent Hardware*, 4(6), 487–505.

Conference Proceedings

- [C26] Catinello, A. S., Dunnhofer, M., Farinella, G. M., Frontoni, E., Furnari, A., Micheloni, C., Paolanti, M., Pietrini, R., Salierno, D., **Stacchio, L.**, & Yaar, A. (2025). Ego and exo views for an object-level human behavior analysis and understanding through tracking in retail spaces. In *International Conference on Image Analysis and Processing* (pp. 376–388). Springer Nature Switzerland.
- [C25] Giordano, M., Berardini, D., Frontoni, E., Zingaretti, P., & **Stacchio, L.** (2025). A modular and efficient framework for the development of large language model-based virtual humans: An educational scenario. In *International Conference on Image Analysis and Processing* (pp. 649–660). Springer Nature Switzerland.
- [C24] Garzarella, S., **Stacchio, L.**, Cascarano, P., De Filippo, A., Cervellati, E., & Marfia, G. (2025). A large reconstruction model driven approach to support humans in digitization of dance visual material into 3D environments. In *CEUR Workshop Proceedings* (Vol. 4072, pp. 47–56). CEUR-WS.
- [C23] Asunis, L., Armandi, V., **Stacchio, L.**, Cascarano, P., Augello, G., Carradori, S., Loretto, A., Meglioraldi, J., Vallasciani, G., & Marfia, G. (2025). CRoCs-XR: A cross-reality collaborative experience for fashion X-commerce via virtual and augmented reality. In *2025 IEEE International Symposium on Mixed and Augmented Reality Adjunct (ISMAR-Adjunct)* (pp. 530–535). IEEE.
- [C22] Armandi, V., Loretto, A., **Stacchio, L.**, Cascarano, P., & Marfia, G. (2025). Multi-modal large language model driven augmented reality situated visualization: The case of wine recognition. In *Proceedings of the Winter Conference on Applications of Computer Vision* (pp. 1313–1322).
- [C21] Garzarella, S., **Stacchio, L.**, Cascarano, P., De Filippo, A., Cervellati, E., & Marfia, G. (2024). Preserving and annotating dance heritage material through deep learning tools: A case study on Rudolf Nureyev. In *CEUR Workshop Proceedings* (Vol. 3810, pp. 88–97). CEUR-WS.
- [C20] Hajahmadi, S.[†], **Stacchio, L.**[†], Giacchè, A., Cascarano, P., & Marfia, G. (2024). Investigating extended reality-powered digital twins for sequential instruction learning: The case of the Rubik’s Cube. In *2024 IEEE International Symposium on Mixed and Augmented Reality (ISMAR)* (pp. 259–268). IEEE.
- [C19] **Stacchio, L.**, Pierdicca, R., Paolanti, M., Zingaretti, P., Frontoni, E., Giovanola, B., & Tiribelli, S. (2024). XRAI-Ethics: Towards a robust ethical analysis framework for extended artificial intelligence. In *2024 IEEE International Symposium on Mixed and Augmented Reality Adjunct (ISMAR-Adjunct)* (pp. 214–219). IEEE.
- [C18] **Stacchio, L.**, Paolanti, M., Spigarelli, F., & Frontoni, E. (2025). Deep Armocromia: A novel dataset for face seasonal color analysis and classification. In *Computer Vision – ECCV 2024 Workshops* (Lecture Notes in Computer Science, Vol. 15623, pp. 352–367). Springer.
- [C17] Balloni, E., **Stacchio, L.**, Gorgoglione, L., Paolanti, M., Pierdicca, R., Mancini, A., Frontoni, E., & Zingaretti, P. (2025). A data-centric module for neural rendering. In *Computer Vision – ECCV 2024 Workshops* (Lecture Notes in Computer Science, Vol. 15628, pp. 312–329). Springer.
- [C16] Vallasciani, G., **Stacchio, L.**, Cascarano, P., & Marfia, G. (2024). CreAIXR: Fostering creativity with generative AI in XR environments. In *2024 IEEE International Conference on Metaverse Computing, Networking, and Applications (MetaCom)* (pp. 1–8). IEEE.
- [C15] **Stacchio, L.**, Balloni, E., Gorgoglione, L., Paolanti, M., Frontoni, E., & Pierdicca, R. (2024). X-NR: Towards an extended reality-driven human evaluation framework for neural-rendering. In *International Conference on Extended Reality* (pp. 305–324). Springer Nature Switzerland.
- [C14] **Stacchio, L.**, Armandi, V., Hajahmadi, S., Donatiello, L., & Marfia, G. (2024). M-AGEW: Empowering outdoor workouts with data-driven augmented reality assistance. In *2024 IEEE International Conference on Artificial Intelligence and eXtended and Virtual Reality (AIxVR)* (pp. 301–305). IEEE.

- [C13] **Stacchio, L.**, Vallasciani, G., Augello, G., Carradori, S., Cascarano, P., & Marfia, G. (2024). WiXaRd: Towards a holistic distributed platform for multi-party and cross-reality WebXR experiences. In *2024 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)* (pp. 264–272). IEEE.
- [C12] **Stacchio, L.**, Di Maria, G., & Marfia, G. (2024). Flying in XR: Bridging desktop applications in extended reality through deep learning. In *2024 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)* (pp. 471–476). IEEE.
- [C11] Asunis, L., Cirina, A., **Stacchio, L.**, & Marfia, G. (2023). HOCTOPUS: An open-source cross-reality tool to augment live-streaming remote classes. In *2023 IEEE International Symposium on Mixed and Augmented Reality Adjunct (ISMAR-Adjunct)* (pp. 29–34). IEEE.
- [C10] Di Maria, G., **Stacchio, L.**, & Marfia, G. (2023). Unity-vrlines: Towards a modular extended reality Unity flight simulator. In *International Conference on Entertainment Computing* (pp. 241–250). Springer.
- [C9] **Stacchio, L.**, Scorolli, C., & Marfia, G. (2023). Evaluating human aesthetic and emotional aspects of 3D generated content through extended reality. In *CEUR Workshop Proceedings* (Vol. 3519, pp. 38–49). CEUR-WS.
- [C8] **Stacchio, L.**, Armandi, V., Donatiello, L., & Marfia, G. (2023). Annholotator: A mixed reality collaborative platform for manufacturing work instruction interaction. In *2023 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)* (pp. 418–424). IEEE.
- [C7] **Stacchio, L.**, Angeli, A., Donatiello, L., Giacchè, A., & Marfia, G. (2022). Rethinking augmented wine recognition. In *2022 IEEE International Symposium on Mixed and Augmented Reality Adjunct (ISMAR-Adjunct)* (pp. 560–565). IEEE.
- [C6] **Stacchio, L.**, Angeli, A., Lisanti, G., & Marfia, G. (2022). Searching for cultural relationships through deep learning models. In *CEUR Workshop Proceedings* (Vol. 3266, pp. 1–11). CEUR-WS.
- [C5] **Stacchio, L.**, Angeli, A., Lisanti, G., & Marfia, G. (2022). Applying deep learning approaches to mixed quantitative-qualitative analyses. In *Proceedings of the 2022 ACM Conference on Information Technology for Social Good* (pp. 161–166). ACM.
- [C4] **Stacchio, L.**, Perlino, M., Vagnoni, U., Sasso, F., Scorolli, C., & Marfia, G. (2022). Who will trust my digital twin? Maybe a clerk in a brick and mortar fashion shop. In *2022 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)* (pp. 814–815). IEEE.
- [C3] **Stacchio, L.**, Angeli, A., & Marfia, G. (2021). Empowering locksmith crafts via mobile augmented reality. In *Proceedings of the Conference on Information Technology for Social Good* (pp. 305–308). ACM.
- [C2] **Stacchio, L.**, Angeli, A., Hajahmadi, S., & Marfia, G. (2021). Revive family photo albums through a collaborative environment exploiting the HoloLens 2. In *2021 IEEE International Symposium on Mixed and Augmented Reality Adjunct (ISMAR-Adjunct)* (pp. 378–383). IEEE.
- [C1] **Stacchio, L.**, Hajahmadi, S., & Marfia, G. (2021). Preserving family album photos with the HoloLens 2. In *2021 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)* (pp. 643–644). IEEE.

Submitted / Under Review

- **Stacchio, L.**, et al. *ELEVATE: Designing Human-Centered GenAI Virtual Tutors for Scalable and Inclusive Education*. Computers and Education: Artificial Intelligence. Under review.
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- **Stacchio, L.**, et al. *[Title to be added]*. Interactive Learning Environments. Under review.
- **Stacchio, L.**, et al. *ATHENA: Exploring the Role of Generative AI-Driven Multimodal Content Generation in Immersive Learning through a Context-Aware Approach*. Computer & Education: X Reality. Under review.
- **Stacchio, L.**, et al. *MIXAMI: User Preference-Aware MIXed reality eXercise Avatar with Fitness-Grounded Multimodal Large Language Model*. IEEE International Symposium on Mixed and Augmented Reality. Under review.
- **Stacchio, L.**, et al. *Mixed Neural Rendering: Interactive Mixed Reality Capture and Curation for Neural 3D Reconstruction*. Virtual Reality. Under review.
- **Stacchio, L.**, Ubaldi, A., Galdelli, A., Mauri, M., Frontoni, E., & Gaggioli, A. *Empathic prompting: Non-verbal context integration for multimodal LLM conversations*. Behaviour & Information Technology. Minor review.
- **Stacchio, L.**, et al. *Affect-Aware Human-Agent Conversations: Towards Empathic LLM Interaction*. Journal of Ambient Intelligence and Humanized Computing. Under review.

- **Stacchio, L.**, et al. *Emotional Prompting: Injecting Real-Time Affective Signals into Large Language Models for Empathic Human-AI Interaction*. 1st Workshop on REpresentation learning from Heterogeneous and Multi-source biomEDical data. Under review.
- **Stacchio, L.**, et al. *RELD: Regularization by Latent Denoising*. IEEE Signal Processing Letters. Minor review. ““